

Customer Churn Prediction — Final Report

NextBridge Summer Internship 2026 — AI/ML Track

What was the goal?

We built a machine learning model that predicts which telecom customers are likely to cancel their service (churn). This lets the business reach out to at-risk customers before they leave.

Which model do we recommend?

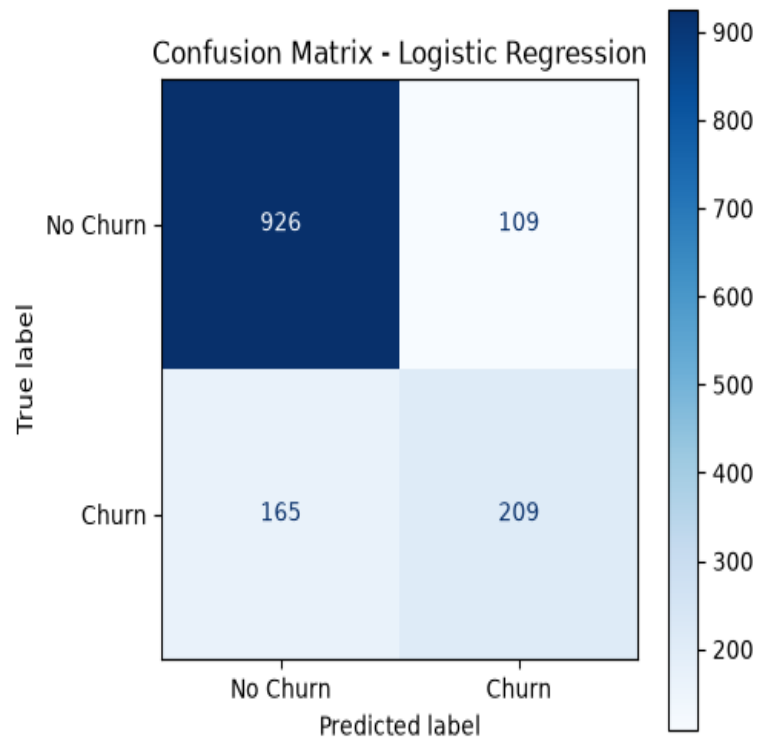
We recommend **Logistic Regression (Baseline)**, which correctly catches **55.9%** of customers who actually churn (this is our chosen priority metric, called recall, because missing a churner costs more than a false alarm). Overall accuracy is 80.6%.

Model Comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression (Baseline)	0.806	0.657	0.559	0.604	0.842
Logistic Regression (Tuned)	0.806	0.657	0.559	0.604	0.842

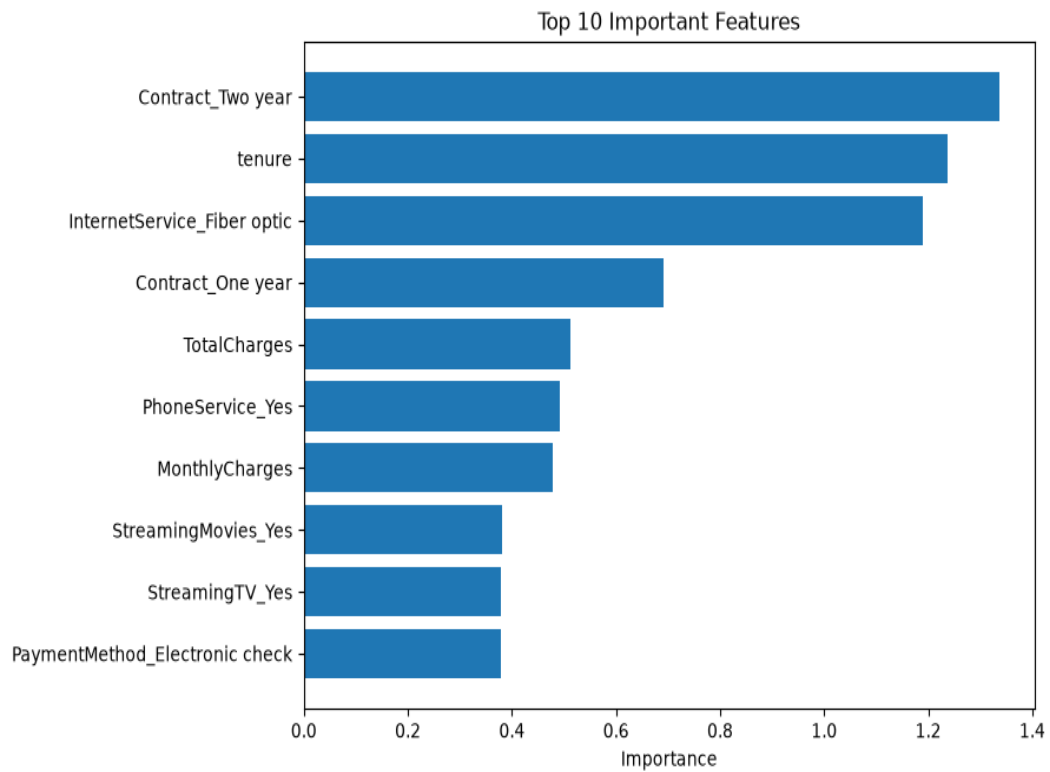
Where does the model make mistakes?

The chart below shows correct vs incorrect predictions on unseen customers (the test set).



What drives churn the most?

The chart below ranks the customer attributes that influenced the model's predictions the most.



Bottom line

The model gives the business an early-warning signal for churn risk, so retention offers can be targeted at the customers most likely to leave, instead of guessing.